

Homework 2 Solution

1. Determine whether each of the following is a linear transformation:

(a) We define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ x_2 \end{pmatrix}, \quad \forall \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2.$$

Solution: T is not linear since

$$T \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

(b) We define $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ by

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 - 2x_2 + x_4 \\ 3x_1 + x_3 - 5x_4 \end{pmatrix}, \quad \forall \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4.$$

Solution: Let

$$A = \begin{pmatrix} 1 & -2 & 0 & 1 \\ 3 & 0 & 1 & -5 \end{pmatrix}.$$

Then for every $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4$,

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = A \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}.$$

Hence $T = T_A$, so T is linear.

(c) We define $T : M_2(\mathbb{R}) \rightarrow \mathbb{R}$ by

$$T(A) = \det(A), \quad \forall A \in M_2(\mathbb{R}).$$

Solution: Note that

$$T(2I_2) = \det(2I_2) = 4 \neq 2 = 2\det(I_2) = 2T(I_2).$$

Therefore T is not a linear transformation.

(d) We define $T : \mathbb{R}^2 \rightarrow \mathbb{C}$ by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = ix + y, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2.$$

Solution: Let $\alpha_1, \alpha_2 \in \mathbb{R}$ and let $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \in \mathbb{R}^2$. Then

$$\begin{aligned} T \left(\alpha_1 \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \alpha_2 \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right) &= T \begin{pmatrix} \alpha_1 x_1 + \alpha_2 x_2 \\ \alpha_1 y_1 + \alpha_2 y_2 \end{pmatrix} = i(\alpha_1 x_1 + \alpha_2 x_2) + (\alpha_1 y_1 + \alpha_2 y_2) \\ &= \alpha_1(ix_1 + y_1) + \alpha_2(ix_2 + y_2) = \alpha_1 T \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \alpha_2 T \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}. \end{aligned}$$

Hence T is linear.

(e) We define $T : \mathbb{R}_2[x] \rightarrow \mathbb{R}_2[x]$ by

$$T(p)(x) = p(x+1) - p(1), \quad \forall p(x) \in \mathbb{R}_2[x].$$

Solution: Let $\alpha_1, \alpha_2 \in \mathbb{R}$ and let $p_1(x), p_2(x) \in \mathbb{R}_2[x]$. Then

$$\begin{aligned} T((\alpha_1 p_1 + \alpha_2 p_2)(x)) &= (\alpha_1 p_1 + \alpha_2 p_2)(x+1) - (\alpha_1 p_1 + \alpha_2 p_2)(1) \\ &= \alpha_1 (p_1(x+1) - p_1(1)) + \alpha_2 (p_2(x+1) - p_2(1)) = \alpha_1 T(p_1)(x) + \alpha_2 T(p_2)(x). \end{aligned}$$

Hence T is linear.

(f) We define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{cases} \begin{pmatrix} x \\ y \end{pmatrix}, & \text{if } x \geq 0, \\ \begin{pmatrix} -x \\ y \end{pmatrix}, & \text{if } x < 0. \end{cases}$$

Solution: Note that

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} |x| \\ y \end{pmatrix}, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2.$$

Then

$$T \begin{pmatrix} -2 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} \neq \begin{pmatrix} -2 \\ -2 \end{pmatrix} = -2 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = -2 \cdot T \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Therefore T is not linear.

2. Prove that each of the following transformations is a linear transformation. Find bases for $\text{Im } T$ and $\ker T$, and determine whether T is surjective and injective.

(a) We define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \\ x+y \end{pmatrix}, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2.$$

Hint. Prove that $T = T_A$.

Solution: For every $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$,

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \\ x+y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Hence $T = T_A$ with $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$, so T is linear.

For the image,

$$\text{Im } T = \left\{ \begin{pmatrix} x \\ y \\ x+y \end{pmatrix} : x, y \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

The two spanning vectors are linearly independent, hence they form a basis for $\text{Im } T$. In particular, $\dim(\text{Im } T) = 2 < 3$, so T is not surjective.

For the kernel,

$$\ker T = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} : \begin{pmatrix} x \\ y \\ x+y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\} = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

Thus $\ker T = \{0\}$, so T is injective (a basis for $\ker T$ is the empty set).

(b) We define $T : \mathbb{R}_3[x] \rightarrow \mathbb{R}_1[x]$ by

$$T(a_0 + a_1x + a_2x^2 + a_3x^3) = a_0 + a_1x, \quad \forall a_0 + a_1x + a_2x^2 + a_3x^3 \in \mathbb{R}_3[x].$$

Solution: Let $\alpha_1, \alpha_2 \in \mathbb{R}$ and let $p_1(x), p_2(x) \in \mathbb{R}_3[x]$. Write

$$p_1(x) = a_0 + a_1x + a_2x^2 + a_3x^3, \quad p_2(x) = b_0 + b_1x + b_2x^2 + b_3x^3.$$

Then

$$\begin{aligned} T(\alpha_1 p_1 + \alpha_2 p_2) &= T((\alpha_1 a_0 + \alpha_2 b_0) + (\alpha_1 a_1 + \alpha_2 b_1)x + (\alpha_1 a_2 + \alpha_2 b_2)x^2 + (\alpha_1 a_3 + \alpha_2 b_3)x^3) \\ &= (\alpha_1 a_0 + \alpha_2 b_0) + (\alpha_1 a_1 + \alpha_2 b_1)x \\ &= \alpha_1(a_0 + a_1x) + \alpha_2(b_0 + b_1x) = \alpha_1 T(p_1) + \alpha_2 T(p_2). \end{aligned}$$

Thus T is linear.

Moreover,

$$\operatorname{Im} T = \{a_0 + a_1x : a_0, a_1 \in \mathbb{R}\} = \mathbb{R}_1[x],$$

so T is surjective and a basis for $\operatorname{Im} T$ is $\{1, x\}$.

Finally,

$$\ker T = \{a_0 + a_1x + a_2x^2 + a_3x^3 : a_0 + a_1x = 0\} = \{a_2x^2 + a_3x^3 : a_2, a_3 \in \mathbb{R}\} = \operatorname{Span}\{x^2, x^3\}.$$

So a basis for $\ker T$ is $\{x^2, x^3\}$, and T is not injective.

3. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear operator, and suppose that

$$T \begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \text{and} \quad T \begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

(a) Compute $T \begin{pmatrix} 0 \\ -5 \end{pmatrix}$.

Solution: We have

$$\begin{pmatrix} 0 \\ -5 \end{pmatrix} = 3 \begin{pmatrix} 2 \\ -3 \end{pmatrix} - 2 \begin{pmatrix} 3 \\ -2 \end{pmatrix},$$

so by linearity,

$$T \begin{pmatrix} 0 \\ -5 \end{pmatrix} = 3T \begin{pmatrix} 2 \\ -3 \end{pmatrix} - 2T \begin{pmatrix} 3 \\ -2 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 2 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}.$$

(b) Prove that T is invertible.

Hint. Use the theorem *basis-to-basis*.

Solution: The vectors $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ are linearly independent in $\mathbb{R}^2$, hence they form a basis of $\mathbb{R}^2$. Their images

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

are also linearly independent, hence they form a basis of $\mathbb{R}^2$. Therefore T maps a basis to a basis, so T is invertible.

(c) Compute $T^{-1}\begin{pmatrix} 4 \\ 6 \end{pmatrix}$.

Solution: First, by item (b), T is invertible and therefore T^{-1} is well-defined. Also, as T is a linear transformation we also know T^{-1} is a linear transformation. We now compute

$$T^{-1}\begin{pmatrix} 4 \\ 6 \end{pmatrix} = T^{-1}\left(2\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = 2T^{-1}\begin{pmatrix} 1 \\ 2 \end{pmatrix} + 2T^{-1}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2\begin{pmatrix} 2 \\ -3 \end{pmatrix} + 2\begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 10 \\ -10 \end{pmatrix}.$$

4. Let $2 \leq n \in \mathbb{N}$ and let $\Delta : M_n(\mathbb{F}) \rightarrow \mathbb{F}$. Suppose that Δ is both linear and multilinear. Prove that $\Delta(A) = 0_{\mathbb{F}}$ for every $A \in M_n(\mathbb{F})$.

Solution: Let $A \in M_n(\mathbb{F})$ and let $R_1, \dots, R_n \in M_{1 \times n}(\mathbb{F})$ be the rows of A , so

$$A = \begin{pmatrix} R_1 \\ \vdots \\ R_n \end{pmatrix}.$$

By multilinearity (linearity in each row), we have

$$\Delta \begin{pmatrix} R_1 + R_1 \\ R_2 \\ \vdots \\ R_n \end{pmatrix} = \Delta(A) + \Delta(A).$$

On the other hand, by linearity of Δ as a map on $M_n(\mathbb{F})$,

$$\Delta \begin{pmatrix} R_1 + R_1 \\ R_2 \\ \vdots \\ R_n \end{pmatrix} = \Delta \begin{pmatrix} R_1 \\ R_2 \\ \vdots \\ R_n \end{pmatrix} + \Delta \begin{pmatrix} R_1 \\ \vec{0}^t \\ \vdots \\ \vec{0}^t \end{pmatrix} = \Delta(A) + \Delta \begin{pmatrix} R_1 \\ \vec{0}^t \\ \vdots \\ \vec{0}^t \end{pmatrix}.$$

Comparing the two expressions gives

$$\Delta(A) + \Delta(A) = \Delta(A) + \Delta \begin{pmatrix} R_1 \\ \vec{0}^t \\ \vdots \\ \vec{0}^t \end{pmatrix} \implies \Delta(A) = \Delta \begin{pmatrix} R_1 \\ \vec{0}^t \\ \vdots \\ \vec{0}^t \end{pmatrix}.$$

Finally, since Δ is multilinear, it is linear in the second row, so

$$\Delta \begin{pmatrix} R_1 \\ \vec{0}^t \\ \vdots \\ \vec{0}^t \end{pmatrix} = \Delta \begin{pmatrix} R_1 \\ 0 \cdot \vec{0}^t \\ \vdots \\ \vec{0}^t \end{pmatrix} = 0 \cdot \Delta \begin{pmatrix} R_1 \\ \vec{0}^t \\ \vdots \\ \vec{0}^t \end{pmatrix} = 0.$$

Therefore $\Delta(A) = 0$ for every $A \in M_n(\mathbb{F})$.

5. Let V, W be two vector spaces over the field F , let $T : V \rightarrow W$ be a linear transformation and let $v_1, \dots, v_n \in V$. Prove or disprove each of the following statements:

(a) There exists a linear operator $S : \mathbb{R}_2[x] \rightarrow \mathbb{R}_2[x]$ for which

$$S(1 + 2x - x^2) = 1 + x, \quad S(2 - x + x^2) = x^2, \quad S(-5x^2 + 7x - 4) = 1 - x + x^2.$$

Solution: False. Assume toward a contradiction that such an S exists. Note that

$$-5x^2 + 7x - 4 = 2(1 + 2x - x^2) - 3(2 - x + x^2).$$

Applying S on both sides of the equation gives

$$1 - x + x^2 = S(-5x^2 + 7x - 4) = S(2(1 + 2x - x^2) - 3(2 - x + x^2))$$

$$\stackrel{\text{linearity}}{=} 2S(1 + 2x - x^2) - 3S(2 - x + x^2)$$

$$= 2(1 + x) - 3x^2$$

$$= 2 + 2x - 3x^2.$$

It follows that $1 - x + x^2 = 2 + 2x - 3x^2$. Contradiction.

(b) If $v_1, \dots, v_n$ span V and if $T(v_1), \dots, T(v_n)$ are linearly independent, then T is invertible.

Solution: False. Consider $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2.$$

Note that for every $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$ we have

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}.$$

Thus, the matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ satisfies $T = T_A$ and therefore T is linear. Choose $\vec{v}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\vec{v}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. These vectors

span $\mathbb{R}^2$ (as they form the standard basis) and

$$T(\vec{v}_1) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad T(\vec{v}_2) = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

It follows that $T(\vec{v}_1), T(\vec{v}_2)$ are linearly independent (as they are part of the standard basis of $\mathbb{R}^3$). But $T(\vec{v}_1), T(\vec{v}_2)$ do not form a basis for $\mathbb{R}^3$, and therefore T doesn't transform a basis to a basis, hence T is not invertible.

(c) If $v_1, \dots, v_n$ are linearly dependent, then $T(v_1), \dots, T(v_n)$ are linearly dependent.

Solution: True. Since $v_1, \dots, v_n$ are linearly dependent, there exist $\alpha_1, \dots, \alpha_n \in F$, not all zero, such that

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0_V.$$

Applying T and using linearity,

$$\alpha_1 T(v_1) + \dots + \alpha_n T(v_n) = T(\alpha_1 v_1 + \dots + \alpha_n v_n) = T(0_V) = 0_W.$$

Thus $T(v_1), \dots, T(v_n)$ are linearly dependent.

6. Let V be a vector space over a field F and let $T : V \rightarrow V$ be a linear operator. Suppose that there exist $0_V \neq v_1, v_2, v_3 \in V$ such that $T(v_1) = v_3$, $T(v_2) = v_2$ and $T(v_3) = -v_1$.

(a) Prove that if $F = \mathbb{R}$ then v_1, v_2, v_3 are linearly independent.

Hint. First prove that v_1, v_3 are linearly independent.

Solution:

We first prove that v_1, v_3 are linearly independent: Assume toward a contradiction that v_1, v_3 are linearly dependent. Since $v_3 \neq 0_V$, there exists an $\alpha \in \mathbb{R}$ such that $v_1 = \alpha \cdot v_3$. Thus

$$v_3 = T(v_1) = T(\alpha \cdot v_3) \stackrel{\text{linearity}}{=} \alpha \cdot T(v_3) = -\alpha \cdot v_1.$$

It follows that

$$v_1 = \alpha \cdot v_3 = -\alpha^2 \cdot v_1$$

and therefore

$$(\alpha^2 + 1) \cdot v_1 = \alpha^2 \cdot v_1 + v_1 = 0_V$$

As $v_1 \neq 0_V$ then $\alpha^2 + 1 = 0$. Contradiction, as the equation $\alpha^2 + 1 = 0$ has no solution in $\mathbb{R}$.

We now prove that v_1, v_2, v_3 are linearly independent: Let $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ such that

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0_V \quad (\star)$$

Applying T to both sides of the equation, we get

$$\alpha_1 T(v_1) + \alpha_2 T(v_2) + \alpha_3 T(v_3) = T(0_V) = 0_V$$

and using the assumption, we get

$$-\alpha_3 v_1 + \alpha_2 v_2 + \alpha_1 v_3 = 0_V$$

Subtracting the two equations yields

$$(\alpha_1 + \alpha_3) v_1 + (\alpha_3 - \alpha_1) v_3 = 0_V$$

As we already know v_1, v_3 are linearly independent, then $\alpha_1 + \alpha_3 = 0$ and $\alpha_3 - \alpha_1 = 0$. Adding these equations together gives $2\alpha_3 = 0$, thus $\alpha_3 = \alpha_1 = 0$. Substituting into $(\star)$ gives $\alpha_2 v_2 = 0_V$ and as $v_2 \neq 0_V$ then $\alpha_2 = 0$.

(b) Give an example where v_1, v_2, v_3 are linearly dependent.

Solution:

Consider the function $id : \mathbb{Z}_2^2 \rightarrow \mathbb{Z}_2^2$ defined by:

$$id \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \forall \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}_2^2.$$

We already know that id is a linear transformation. We now choose

$$\vec{v}_1 = \vec{v}_2 = \vec{v}_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Note that

$$id(\vec{v}_1) = \vec{v}_1 = \vec{v}_3, \quad id(\vec{v}_2) = \vec{v}_2, \quad id(\vec{v}_3) = \vec{v}_3 = \vec{v}_1 = -\vec{v}_1$$

and clearly, $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are linearly dependent.